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Assignment No: 03

Practical Title: Creating an Application in Salesforce.com using Apex programming Language.

Objectives:

- ☐ To understand the basics of Salesforce platform
- ☐ To learn Apex programming language
- ☐ To create a simple application in Salesforce
- ☐ To execute Apex code and understand its working

Software Requirement:

- ☐ Web Browser (Chrome/Edge)
- ☐ Salesforce Developer Account
- ☐ Internet Connection

Theory:

1. Salesforce:-

Salesforce is a **cloud-based Customer Relationship Management (CRM) platform** that helps businesses manage their customer data, sales activities, marketing processes, and customer support services in one centralized system.

Salesforce allows organizations to store customer information such as contact details, communication history, purchase records, and service requests. This helps companies build better relationships with customers and improve business efficiency.

It operates on a **cloud computing model**, which means users do not need to install any software. They can access Salesforce through a web browser from anywhere, making it highly flexible and scalable.

Salesforce provides various tools and services, including:

- **Sales Cloud** – for managing sales processes and leads
- **Service Cloud** – for customer support and case management
- **Marketing Cloud** – for digital marketing and campaigns
- **Commerce Cloud** – for e-commerce solutions

2. Difference Between Salesforce and Amazon AWS:-

Feature / Aspect	Salesforce	AWS (Amazon Web Services)
Primary Purpose	Customer Relationship Management (CRM) platform	Cloud computing platform
Focus Area	Sales, marketing, customer service, business apps	Infrastructure, storage, computing, networking
Type of Service	SaaS (Software as a Service)	IaaS, PaaS, SaaS
Main Users	Sales teams, marketers, customer support teams	Developers, IT professionals, enterprises
Core Functionality	Manage customer data, automate sales, analytics	Host applications, run servers, databases, AI services
Customization	High (via Apex, Lightning, workflows)	High (via APIs, SDKs, services configuration)
Programming Languages	Apex (proprietary), SOQL	Supports many (Python, Java, Node.js, etc.)
Deployment	Fully cloud-based (no infrastructure needed)	Cloud-based with full infrastructure control
Examples of Services	Sales Cloud, Service Cloud, Marketing Cloud	EC2, S3, Lambda, RDS
Ease of Use	More user-friendly (non-technical users too)	More technical (requires cloud knowledge)
Scalability	Scales CRM operations	Highly scalable infrastructure globally
Pricing Model	Subscription-based (per user/month)	Pay-as-you-go (usage-based)
Best For	Managing customer relationships and sales pipelines	Building and deploying applications & systems
Learning Curve	Moderate (business-focused)	Steeper (technical concepts involved)

3. Apex Language:-

Apex is a **strongly typed, object-oriented programming language** developed by Salesforce, used to build custom business logic and applications on the Salesforce platform.

Apex is similar to Java in syntax and structure, which makes it easy for developers to learn. It is specifically designed to execute flow and transaction control statements on Salesforce servers. This means that all the code written in Apex runs on the cloud rather than on a local machine.

Apex is mainly used to:

- Perform database operations such as inserting, updating, deleting, and retrieving records
- Create business logic and automation
- Develop triggers that execute automatically when data changes
- Build custom controllers for user interfaces

One of the key features of Apex is its ability to work directly with Salesforce databases using SOQL (Salesforce Object Query Language). It also supports exception handling, loops, conditional statements, and other programming concepts.

4. Features of Apex:-

- Runs on Salesforce servers
- Supports object-oriented concepts
- Easy integration with database (SOQL)
- Similar to Java syntax

5. Salesforce Architecture:-

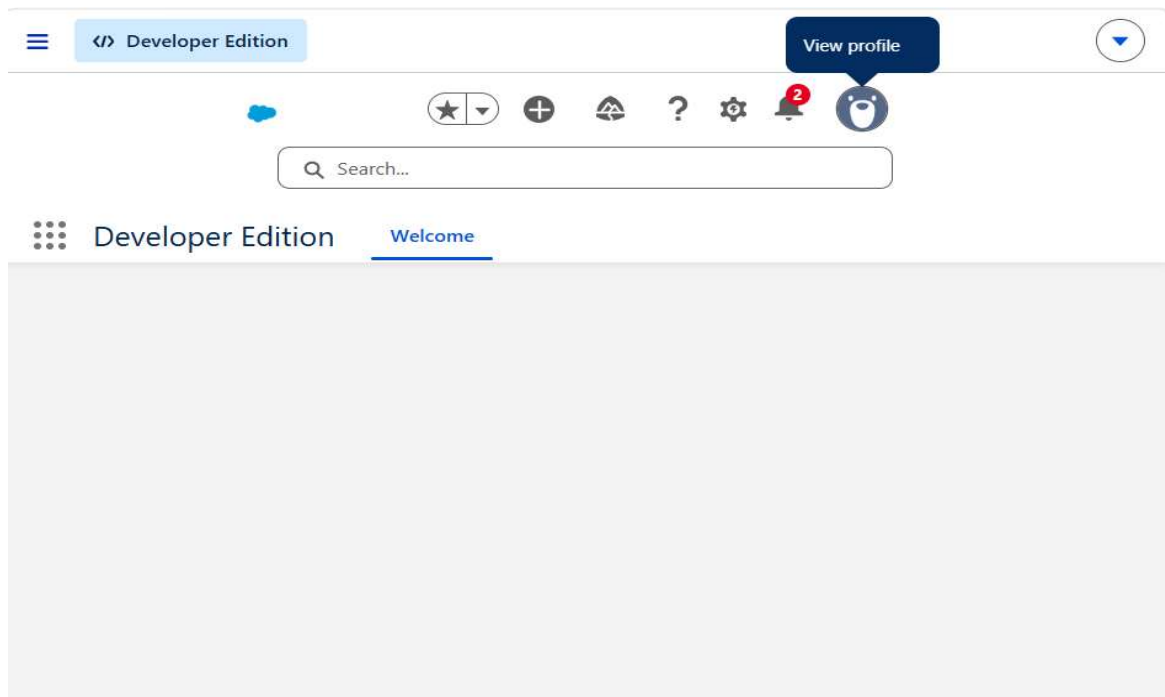
Salesforce uses a multi-tenant architecture where multiple users share the same resources securely.

6. Steps to Create Application using Apex:-

a. Login to Salesforce :-

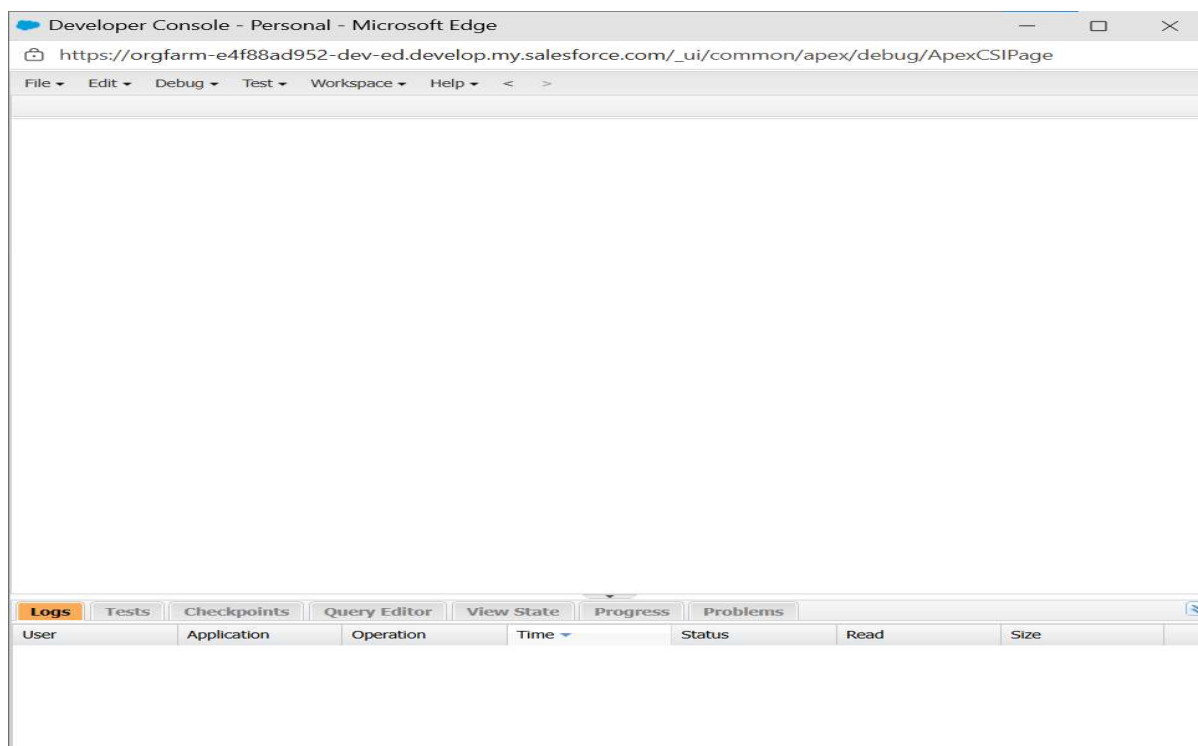
- Go to Salesforce Developer account

- Login using credentials



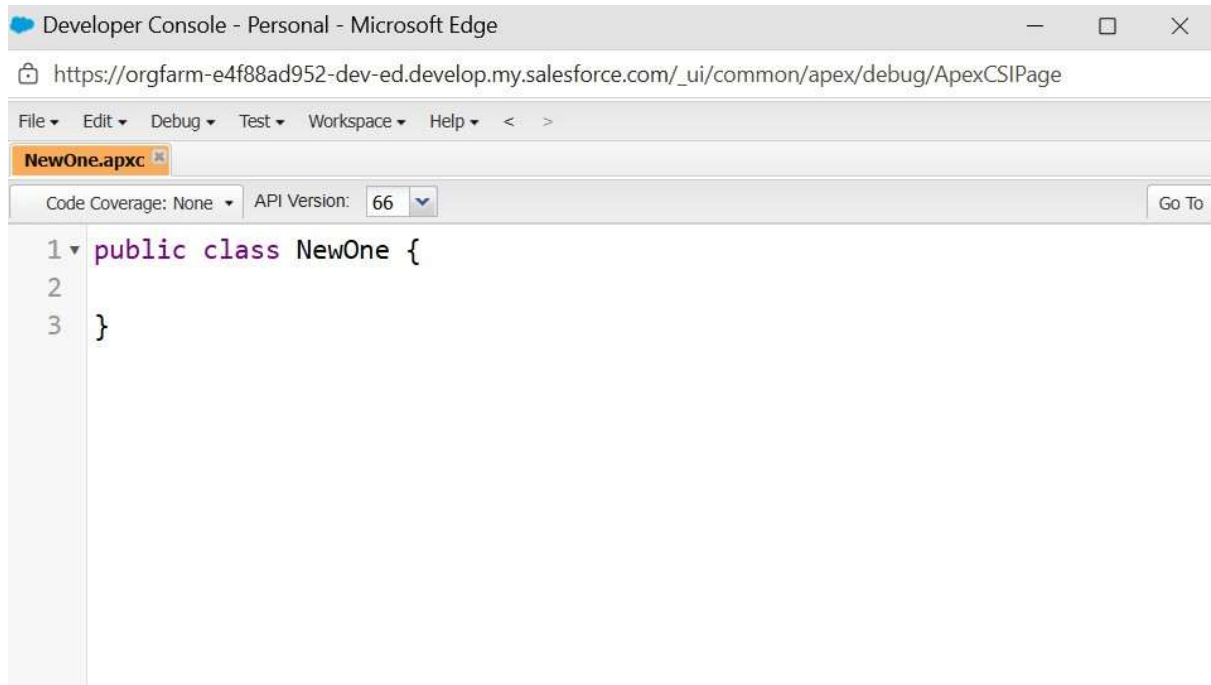
b. Open Developer Console :-

- Click on profile icon
- Select **Developer Console**

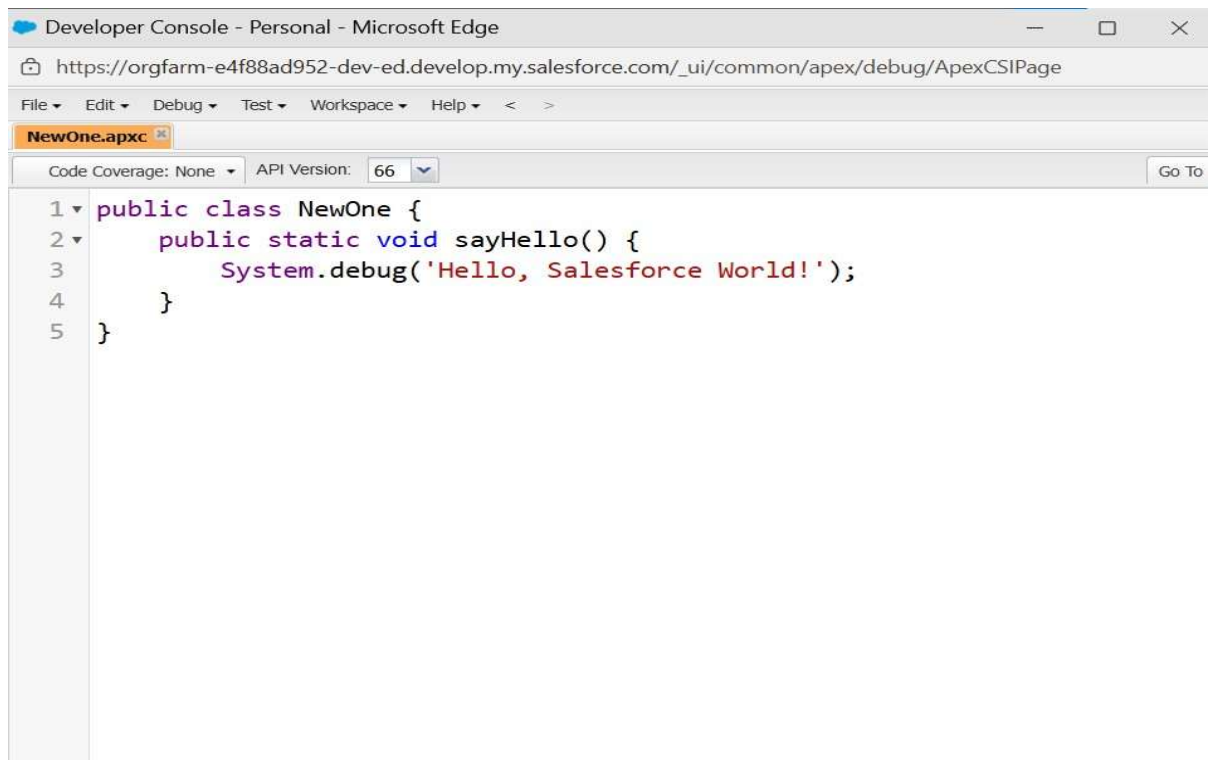


c. Create Apex Class :-

- Go to **File** → **New** → **Apex Class**
- Name it (e.g., NewOne)



d. Write Apex Code :-



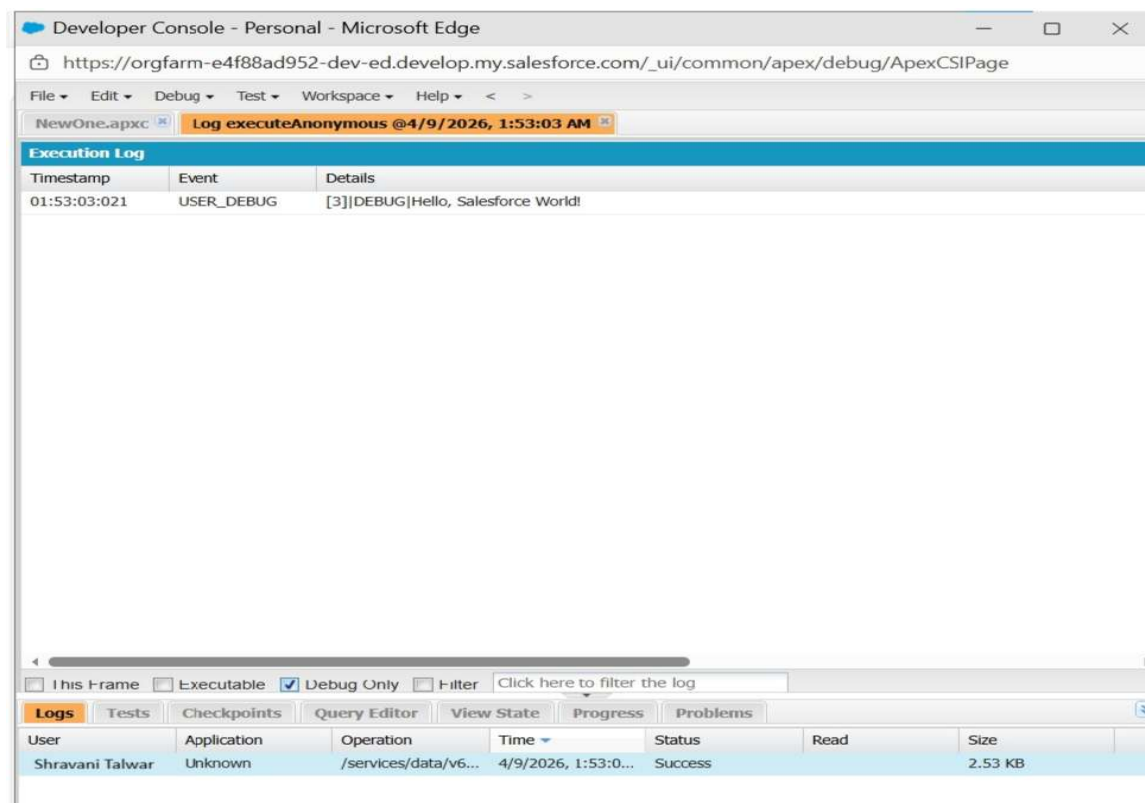
e. Execute Code :-

- Go to **Debug** → **Open Execute Anonymous Window**

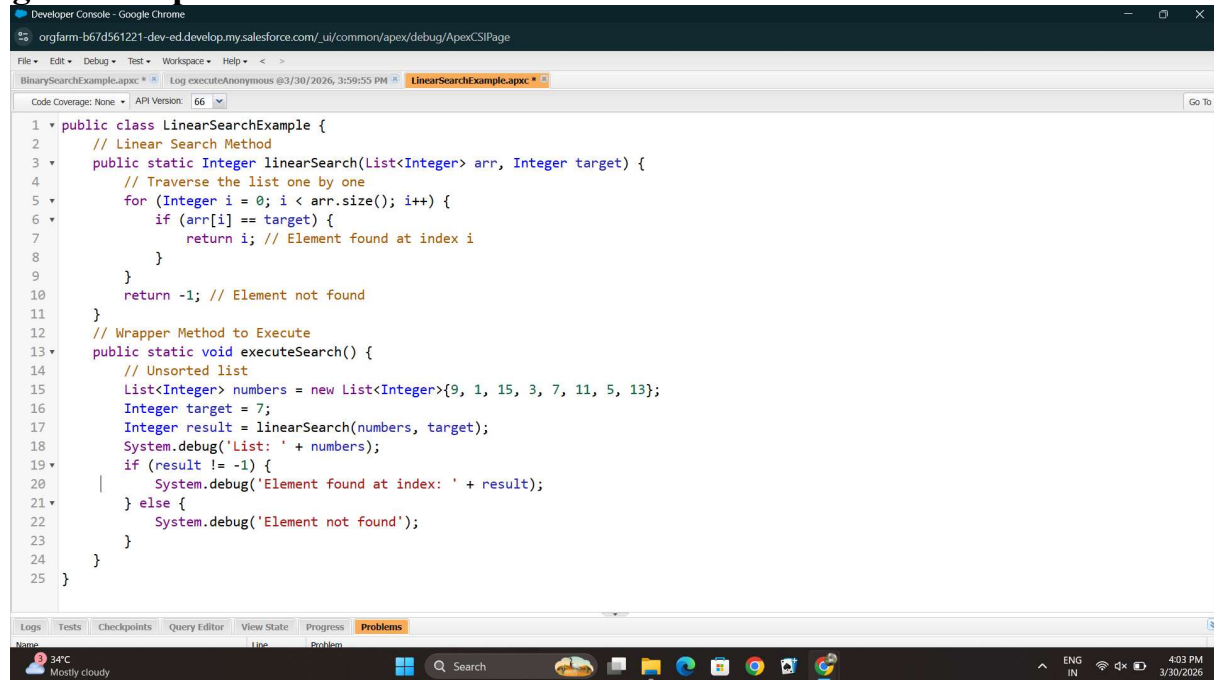


f. View Output :-

- Check **Logs** tab
- Output will be:



g. Write Apex Code for Linear Search :-

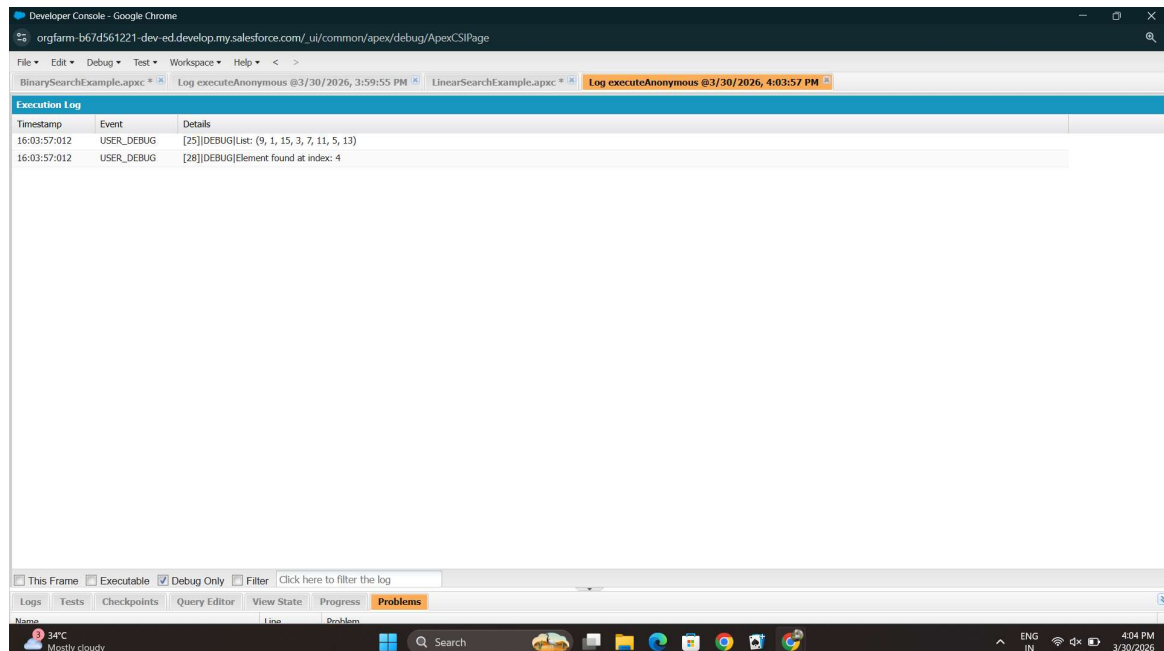


The screenshot shows the Salesforce Developer Console with the Apex code for a linear search algorithm. The code is as follows:

```
1 public class LinearSearchExample {
2     // Linear Search Method
3     public static Integer linearSearch(List<Integer> arr, Integer target) {
4         // Traverse the list one by one
5         for (Integer i = 0; i < arr.size(); i++) {
6             if (arr[i] == target) {
7                 return i; // Element found at index i
8             }
9         }
10        return -1; // Element not found
11    }
12    // Wrapper Method to Execute
13    public static void executeSearch() {
14        // Unsorted list
15        List<Integer> numbers = new List<Integer>{9, 1, 15, 3, 7, 11, 5, 13};
16        Integer target = 7;
17        Integer result = linearSearch(numbers, target);
18        System.debug('List: ' + numbers);
19        if (result != -1) {
20            System.debug('Element found at index: ' + result);
21        } else {
22            System.debug('Element not found');
23        }
24    }
25 }
```

h. View Output :-

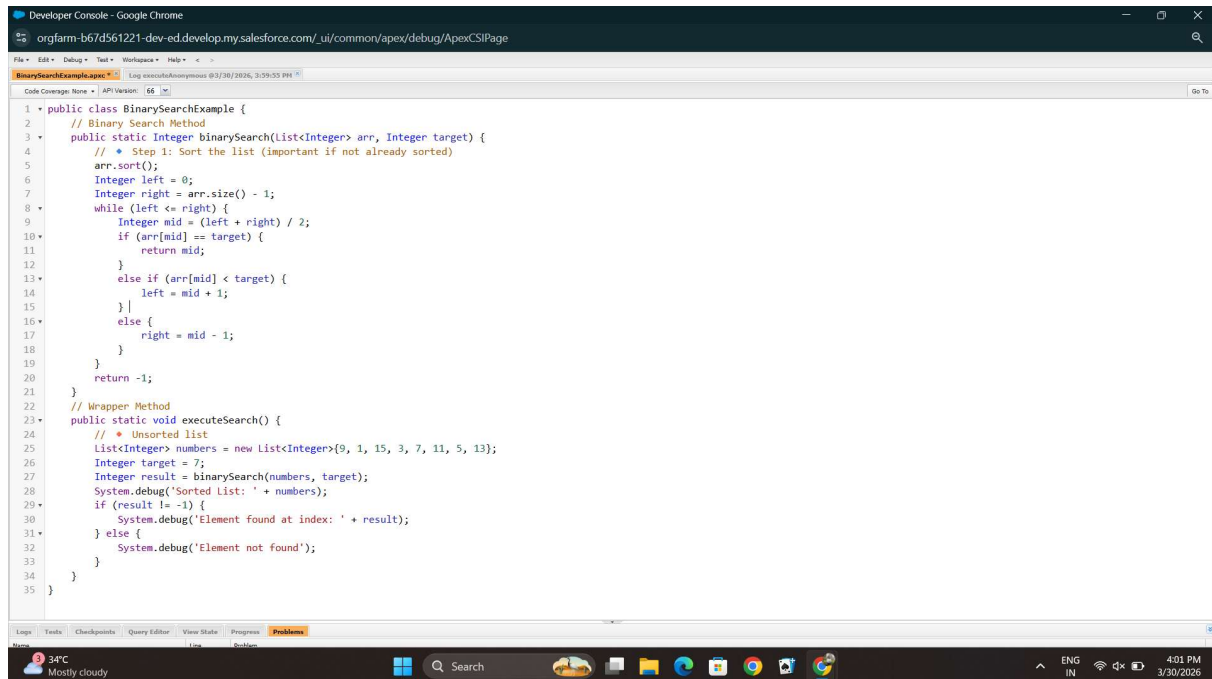
- Go to Debug → Open Execute Anonymous Window
- Check **Logs** tab
- Output will be:



The screenshot shows the Salesforce Developer Console with the Execution Log tab selected. The log contains the following entries:

Timestamp	Event	Details
16:03:57:012	USER_DEBUG	[25]DEBUG List: (9, 1, 15, 3, 7, 11, 5, 13)
16:03:57:012	USER_DEBUG	[28]DEBUG Element found at index: 4

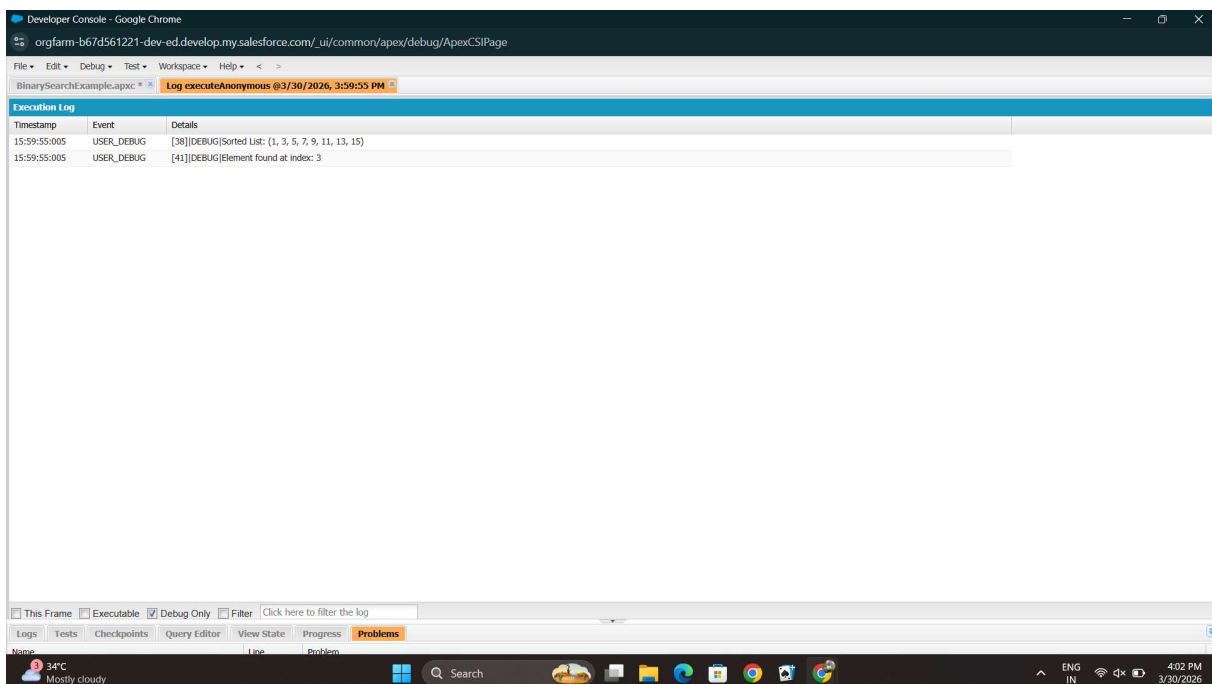
i. Write Apex Code for Binary Search :-



```
1 public class BinarySearchExample {
2     // Binary Search Method
3     public static Integer binarySearch(List<Integer> arr, Integer target) {
4         // * Step 1: Sort the list (important if not already sorted)
5         arr.sort();
6         Integer left = 0;
7         Integer right = arr.size() - 1;
8         while (left <= right) {
9             Integer mid = (left + right) / 2;
10            if (arr[mid] == target) {
11                return mid;
12            }
13            else if (arr[mid] < target) {
14                left = mid + 1;
15            }
16            else {
17                right = mid - 1;
18            }
19        }
20        return -1;
21    }
22    // Wrapper Method
23    public static void executeSearch() {
24        // * Unsorted list
25        List<Integer> numbers = new List<Integer>{9, 1, 15, 3, 7, 11, 5, 13};
26        Integer target = 7;
27        Integer result = binarySearch(numbers, target);
28        System.debug('Sorted List: ' + numbers);
29        if (result != -1) {
30            System.debug('Element found at index: ' + result);
31        } else {
32            System.debug('Element not found');
33        }
34    }
35 }
```

j. Execute Code and View Output :-

- Go to Debug → Open Execute Anonymous Window
- Check **Logs** tab
- Output will be:



Timestamp	Event	Details
15:59:55:005	USER_DEBUG	[38][DEBUG]Sorted List: (1, 3, 5, 7, 9, 11, 13, 15)
15:59:55:005	USER_DEBUG	[41][DEBUG]Element found at index: 3

Conclusion :-

Successfully created a simple application using Apex programming in Salesforce and understood how to execute and debug code.